

ID2221 Data-Intensive Computing

Lab assignment Week 1

Benchmark Report

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Introduction

In this report, we evaluate the performance and scalability of our storage architecture by implementing, testing, and comparing two distinct storage strategies for the integrated Taxi Trips dataset.

Testing Outline & Storage Strategies

We benchmarked two storage strategies, one without any partitioning schemes (strategy A) and one with partitioning by pickup month on the integrated Taxi Trips dataset (strategy B).

To measure system performance, we tracked four key storage metrics: ingestion time (seconds), storage size (MB), query latency (seconds) and number of generated files.

We calculate the ingestion time in our benchmarking program by comparing the start and end times before and after we ingest the dataset and create the Delta tables. The size and file count are calculated in our program by iterating through each parquet file in the Delta table and returning the count of files and the sum of the file sizes. The query latency measuring process is outlined below.

Testing Queries

The benchmarking suite evaluates query latency using the following queries:

- **Query 1:** The number of taxi trips per borough.
- **Query 2:** The average trip duration per day.
- **Query 3:** The average fare per borough.

The query latency measured is the total elapsed time for all three queries combined, calculated in our program by comparing the start and end times before and after running the three queries.

Benchmark Results & Performance Discussion

Table 1 presents the results of the benchmarking process across the four metrics defined above.

	Strategy A (Unpartitioned)	Strategy B (Date Partitioned)
Ingestion Time	10.16s	6.67s
Size	376.54 MB	384.1 MB
Files	46	121
Query Latency	1.09s	0.85s

Table 1: Ingestion time, total size, number of files, and query latency associated with the two storage strategies: unpartitioned and date partitioned

The results show that the date-partitioned strategy performs better in terms of both ingestion time and query latency. The ingestion time for the date partitioned strategy is 3.49 seconds lower (down 34%) than the unpartitioned, and the query latency is 0.24 seconds lower (down 22%). On the other hand, partitioning by date results in a slightly larger size and significantly larger number of files. The size is 7.56MB larger (up 1.97%) for the date-partitioned than the unpartitioned, as well as there are 75 more files (up 62%). This clearly illustrates the trade-off between query performance and storage overhead introduced by partitioning. For the purposes of this project, the date-partitioned strategy is preferable since it provides a significant benefit in ingestion time and query latency.